

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & LIVE-I(A)]

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	C	D	C		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B,C	A,C	A,C,D	A,C	A,D	B,C,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	1.20	48.00	4.00	1.00	5.00	2.50
SECTION-II (ii)	Q.	7	8	9			
	A.	94	5	2			

PART-2 : CHEMISTRY

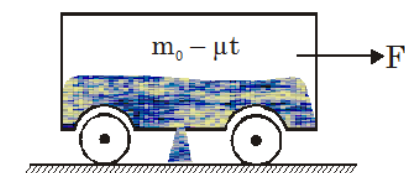
SECTION-I (i)	Q.	1	2	3	4		
	A.	D	D	B	D		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B,C,D	A,B,C	A,B,D	A,B	A	A,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	8.00	7.00	5000.00	1500.00	1.50	0.25
SECTION-II (ii)	Q.	7	8	9			
	A.	7	3	2			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	C	D	D		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B,D	C,D	A,D	A,C,D	A,B,C,D	B,C,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	0.00	0.00	1.00	0.25	3.00	0.33 or 0.34
SECTION-II (ii)	Q.	7	8	9			
	A.	1	14	2			

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

1. Ans (A)

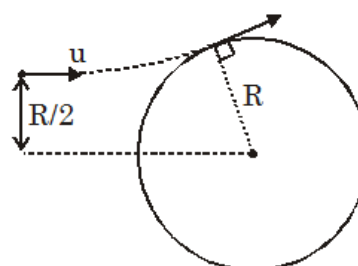


$$F = (m_0 - \mu t)a$$

$$F = (m_0 - \mu t) \frac{dv}{dt}$$

$$\int_0^v dv = \int_0^t \frac{F}{(m_0 - \mu t)} \cdot dt$$

2. Ans (C)



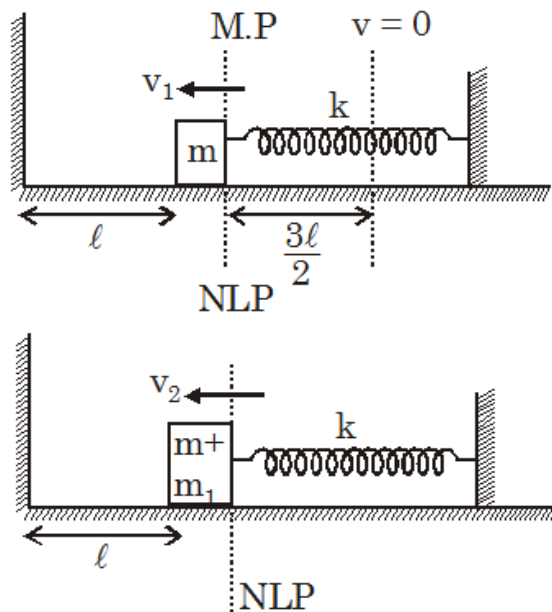
Angular momentum conservation

$$mu \frac{R}{2} = mvR \dots(i)$$

Energy conservation

$$\frac{1}{2} mu^2 + 0 = \frac{1}{2} mv^2 + \frac{kQq}{R}$$

3. Ans (D)

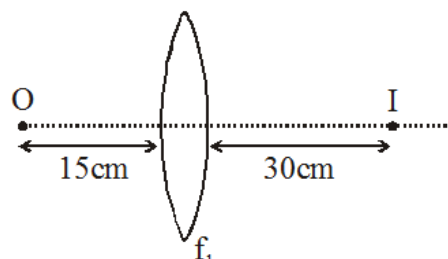


$$v_1 = \omega_1 A_1 = \sqrt{\frac{k}{m} \cdot \frac{3l}{2}}$$

$$v_2 = \omega_2 A_2 = \sqrt{\frac{k}{m+m_1} \cdot l}$$

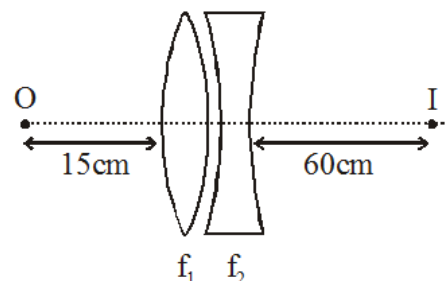
$$mv_1 = (m+m_1)v_2$$

4. Ans (C)



$$\frac{1}{30} - \frac{1}{-15} = \frac{1}{f_1}$$

$$f_1 = 10 \text{ cm}$$



$$\frac{1}{60} - \frac{1}{-15} = \frac{1}{f_{eq}}$$

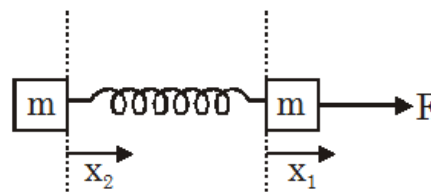
$$\text{and } \frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$\frac{1}{12} = \frac{1}{10} + \frac{1}{f_2}$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (B,C)



$$F = (m+m) a_{cm}$$

$$a_{cm} = \frac{F}{2m} \quad \dots(i)$$

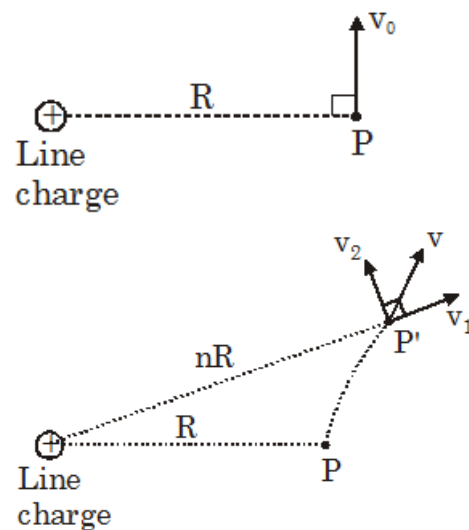
$$S_{cm} = ut + \frac{1}{2}at^2$$

$$\frac{m \cdot x_1 + m x_2}{2m} = 0 + \frac{1}{2} \cdot \frac{F}{2m} \cdot t^2$$

$$\frac{x_1 + x_2}{2} = \frac{F}{4m} t^2 \quad \dots(ii)$$

$$x_1 - x_2 = \ell \quad \dots(iii)$$

6. Ans (A,C)



From energy conservation

$$\frac{1}{2}mv_0^2 + Qv_p = \frac{1}{2}mv^2 + Qv_p'$$

$$\frac{1}{2}mv^2 = \frac{1}{2}mv_0^2 + Q[v_p - v_p']$$

$$= \frac{1}{2}mv_0^2 + Q \left[2k\lambda \cdot \ln \left(\frac{\eta R}{R} \right) \right]$$

$$v = \sqrt{v_0^2 + \frac{4k\lambda \cdot \ln \eta}{m}}$$

From angular momentum conservation

$$mv_0 R = v_2 \eta R$$

$$v_2 = \frac{v_0}{\eta}$$

$$\text{and } v_1^2 + v_2^2 = v^2 \Rightarrow v_1 = \sqrt{v^2 - v_2^2}$$

7. Ans (A,C,D)

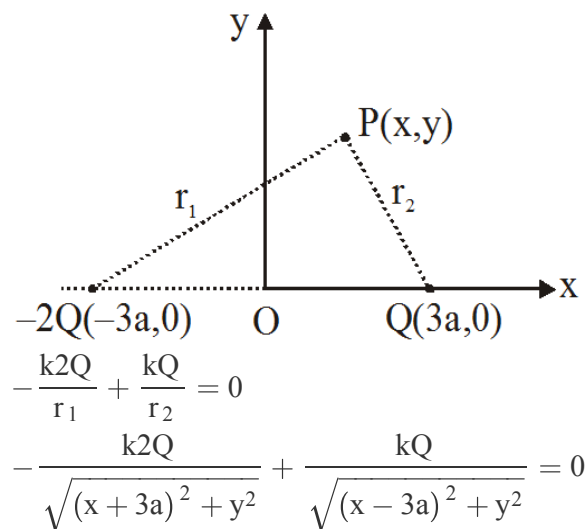
$$T = 2\pi \sqrt{\frac{I}{pE}}$$

$$= 2\pi \sqrt{\frac{\left(\frac{m\ell^2}{4}\right) \times 2}{q\ell E}}$$

In uniform $\vec{E}$ net force of dipole is always zero.

And when $\vec{p}$ is parallel to $\vec{E}$, $\vec{\tau} = \vec{p} \times \vec{E} = 0$

8. Ans (A,C)



$$x^2 + y^2 - 10ax + 9a^2 = 0$$

$$(x - 5a)^2 + y^2 = (4a)^2$$

Radius of circle = $4a$

Centre of circle = $(5a, 0)$

9. Ans (A,D)

$$T_0 = 2\pi \sqrt{\frac{M}{k}}$$

$$T = 2\pi \sqrt{\frac{M+m}{k}} \quad \therefore \frac{T}{T_0} = \sqrt{\frac{M+m}{M}}$$

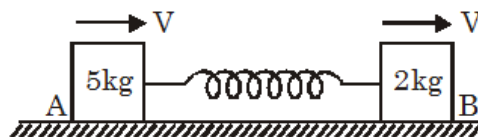
Momentum Conservation

$$Mv_1 = (M+m)v_2$$

$$M.a_0\omega_1 = (M+m)a.\omega_2$$

$$Ma_0\sqrt{\frac{k}{M}} = (M+m)a.\sqrt{\frac{k}{M+m}}$$

10. Ans (B,C,D)



From momentum conservation

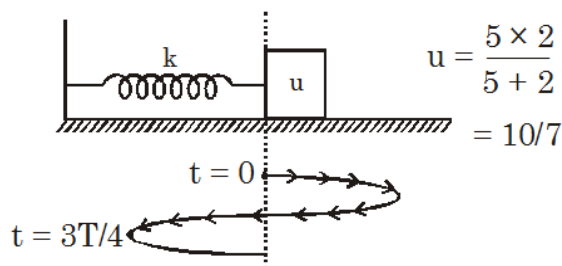
$$3 \times 5 + 2 \times 10 = 7V$$

$$V = 5 \text{ m/s}$$

From energy conservation

$$\frac{1}{2} \times 5 \times 3^2 + \frac{1}{2} \times 2 \times 10^2$$

$$= \frac{1}{2} \times 7 \times 5^2 + \frac{1}{2} \times 1120 \times x^2$$



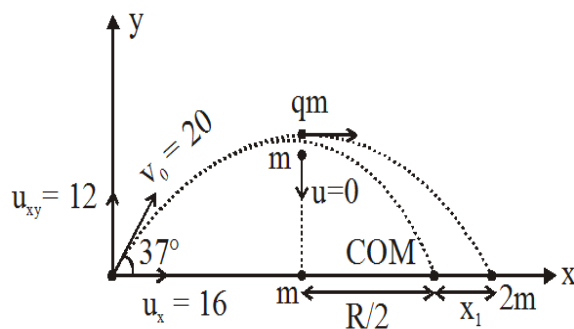
First maximum compression occurs at

$$t = \frac{3T}{4} = \frac{3}{4} \times 2\pi \sqrt{\frac{\mu}{k}}$$

PART-1 : PHYSICS

SECTION-II (i)

1. Ans (1.20)



$$t = \frac{T}{2} = \frac{1}{2} \cdot \frac{2u_y}{g}$$

$$t = \frac{12}{10} = 1.2 \text{ sec}$$

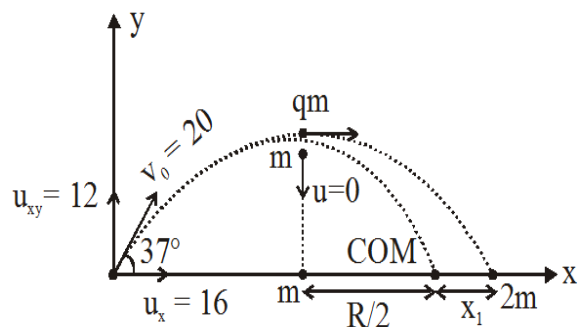
$$m \cdot \frac{R}{2} = 2m \cdot x_1$$

$$x_1 = \frac{R}{4} \quad \therefore x = R + x_1 = \frac{5R}{4}$$

$$x = \frac{5}{4} \times 2 \times \frac{u_x u_y}{g}$$

$$= \frac{5}{4} \times 2 \times \frac{16 \times 12}{10} = 48 \text{ m}$$

2. Ans (48.00)



$$t = \frac{T}{2} = \frac{1}{2} \cdot \frac{2u_y}{g}$$

$$t = \frac{12}{10} = 1.2 \text{ sec}$$

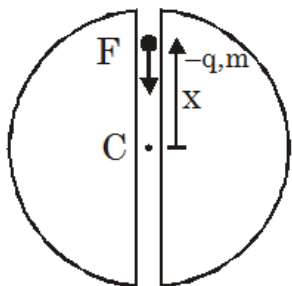
$$m \cdot \frac{R}{2} = 2m \cdot x_1$$

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$$x = \frac{5}{4} \times 2 \times \frac{u_x u_y}{g}$$

$$= \frac{5}{4} \times 2 \times \frac{16 \times 12}{10} = 48 \text{ m}$$

3. Ans (4.00)



$$F = qE = q \cdot \frac{kQ}{R^3} x$$

$$ma = \frac{kQq}{R^3} x$$

$$a = - \left(\frac{Qq}{4\pi\epsilon_0 m R^3} \right) x$$

$\Rightarrow$ SHM with MP at the centre of sphere

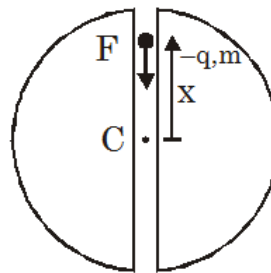
$$\omega = \sqrt{\frac{Qq}{4\pi\epsilon_0 m R^3}}$$

$$T = 2\pi \sqrt{\frac{4\pi\epsilon_0 m R^3}{Qq}}$$

$$\therefore t = \frac{T}{4} = \sqrt{\frac{\pi\epsilon_0 m R^3}{Qq}}$$

$$v = v_{\max} = A\omega = R \sqrt{\frac{Qq}{4\pi\epsilon_0 m R^3}}$$

4. Ans (1.00)



$$F = qE = q \cdot \frac{kQ}{R^3} x$$

$$ma = \frac{kQq}{R^3} x$$

$$a = - \left(\frac{Qq}{4\pi\epsilon_0 m R^3} \right) x$$

$\Rightarrow$ SHM with MP at the centre of sphere

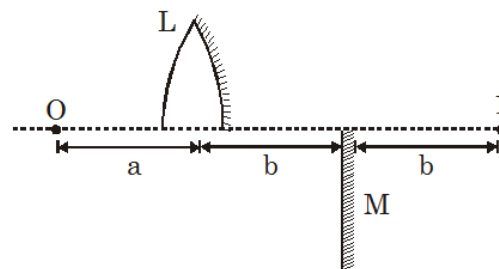
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$$\therefore t = \frac{T}{4} = \sqrt{\frac{\pi\epsilon_0 m R^3}{Qq}}$$

$$v = v_{\max} = A\omega = R \sqrt{\frac{Qq}{4\pi\epsilon_0 m R^3}}$$

5. Ans (5.00)



Power of silvered lens

$$P = \left(\frac{1.5 - 1}{40} \right) \times 2 + \left(\frac{-2 \times 1.5}{-40} \right)$$

$$-\frac{1}{f} = \frac{1}{40} + \frac{3}{40}$$

$$f = -10 \text{ cm}$$

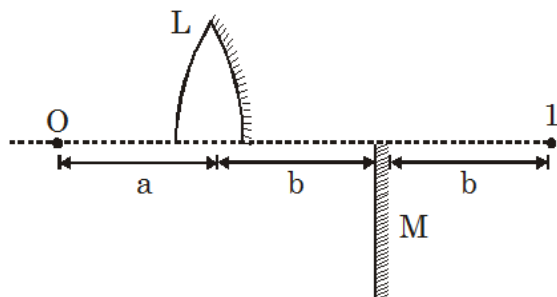
$$\text{For silvered lens } \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{(a + 2b)} + \frac{1}{-a} = \frac{1}{-10} \quad \dots(i)$$

$$\text{And } m = -\frac{v}{u}$$

$$2 = -\frac{a + 2b}{(-a)} \quad \dots(ii)$$

6. Ans (2.50)



Power of silvered lens

$$P = \left(\frac{1.5 - 1}{40} \right) \times 2 + \left(\frac{-2 \times 1.5}{-40} \right)$$

$$-\frac{1}{f} = \frac{1}{40} + \frac{3}{40}$$

$$f = -10 \text{ cm}$$

$$\text{For silvered lens } \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{(a + 2b)} + \frac{1}{-a} = \frac{1}{-10} \quad \dots(i)$$

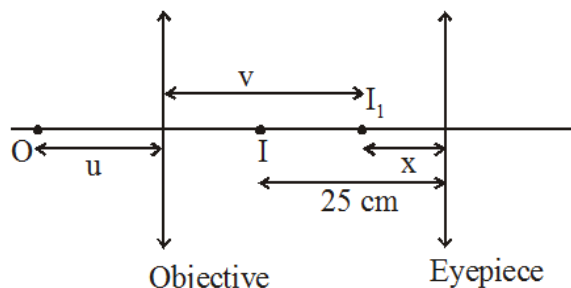
$$\text{And } m = -\frac{v}{u}$$

$$2 = -\frac{a + 2b}{(-a)} \quad \dots(ii)$$

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (94)



$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \text{For eye piece}$$

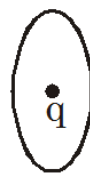
$$\frac{1}{-25} - \frac{1}{-x} = \frac{1}{5}$$

$$\text{and } v = 20 - x$$

$$\text{For objective } \Rightarrow \frac{1}{v} - \frac{1}{u} = \frac{1}{f_0}$$

$$m = \frac{v}{u} \left(1 - \frac{D}{f_e} \right)$$

9. Ans (2)



$$-m\omega^2 x = -\frac{kQqx}{(x^2 + R^2)^{3/2}}$$

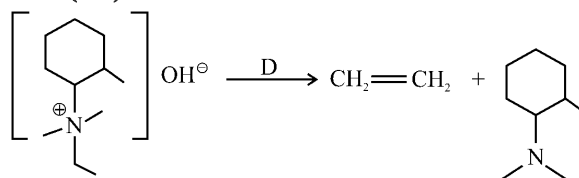
$$\omega = \sqrt{\frac{kQq}{mR^3}} = 16$$

$$\omega' = \sqrt{\frac{kQq}{m(4R)^3}} = \frac{16}{8} = 2$$

PART-2 : CHEMISTRY

SECTION-I (i)

2. Ans (D)



3. Ans (B)

$$\text{Let v.p. of A} = P_A^0 \text{ \& v.p. of B} = P_B^0$$

$$\text{when } \eta_A = 4 ; \eta_B = 6$$

$$k_A = 0.4 ; \eta_B = 0.6$$

$$\text{so } 460 = 0.4 \times P_A^0 + 0.6 \times P_B^0$$

$$\Rightarrow 2P_A^0 + 3P_B^0 = 2300$$

$$\text{now when } \eta_A = 16 \text{ then}$$

$$k_A = \frac{4}{20} \text{ \& } k_B = \frac{16}{20} = \frac{1}{5} ; = \frac{4}{5}$$

$$\text{so } 480 = \frac{1}{5} P_A^0 + \frac{4}{5} P_B^0$$

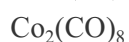
$$P_A^0 + 4P_B^0 = 2400 \dots\dots (1)$$

$$2P_A^0 + 3P_B^0 = 2300 \dots\dots (2)$$

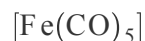
$$\text{solving (1) \& (2) we get } P_A^0 = 400 \text{ \& }$$

$$P_B^0 = 500$$

4. Ans (D)



$$\text{EAN of Co} = 36$$



$$\text{EAN of Fe} = 36$$



$$\text{EAN of V} = 36$$



$$\text{EAN of Mn} = 37$$

PART-2 : CHEMISTRY

SECTION-I (ii)

9. **Ans (A)**

Because Cs^+ has large size as compare to Mg^{+2} , Ca^{+2} and Pb^{+2}

10. **Ans (A,D)**

(i) oxidation state & co-ordination number in both complexes is +3 and 6 respectively

(ii) Number of Co-N linkages in both complexes are 4

PART-2 : CHEMISTRY

SECTION-II (i)

3. **Ans (5000.00)**

reaction is zero order

$$\text{now } -\frac{1}{4} \frac{d[A]}{dt} = K_{\text{reaction}}$$

$$\text{so } -\frac{d[A]}{dt} = 4 \times K = 20 \text{ mm /sec}$$

$$\text{now at} = A_0 - K \times t$$

$$p_A = 4000 - 20 \times 100$$

$$p_A = 2000 \text{ mm of Hg}$$

$$\text{now } p_B = \frac{3}{4} \times 500 = 1500.00$$

$$\text{now } T_{\text{final}} = P_A = 0 \text{ so}$$

$$t = 200 \text{ sec.}$$

so after

$$250 \text{ sec, } P_T = \frac{3}{4} \times 400 + \frac{2}{4} \times 4000$$

$$= 3000 + 2000 = 5000$$

4. **Ans (1500.00)**

reaction is zero order

$$\text{now } -\frac{1}{4} \frac{d[A]}{dt} = K_{\text{reaction}}$$

$$\text{so } -\frac{d[A]}{dt} = 4 \times K = 20 \text{ mm /sec}$$

$$\text{now at} = A_0 - K \times t$$

$$p_A = 4000 - 20 \times 100$$

$$p_A = 2000 \text{ mm of Hg}$$

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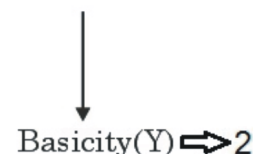
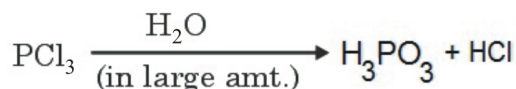
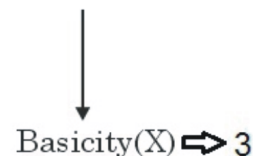
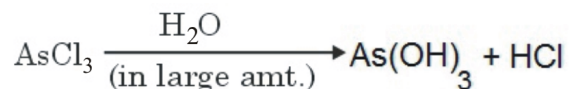
$$t = 200 \text{ sec.}$$

so after

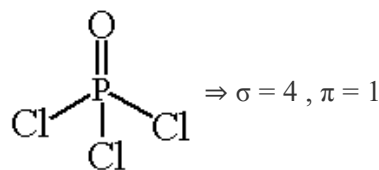
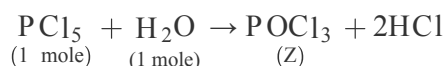
$$250 \text{ sec, } P_T = \frac{3}{4} \times 400 + \frac{2}{4} \times 4000$$

$$= 3000 + 2000 = 5000$$

5. **Ans (1.50)**



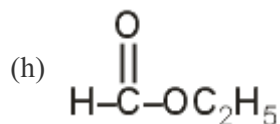
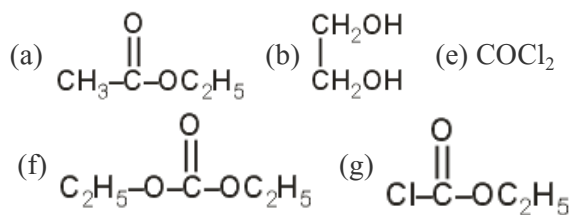
6. **Ans (0.25)**



PART-2 : CHEMISTRY

SECTION-II (ii)

7. Ans (7)



(i) $\text{C}_6\text{H}_5\text{COCl}$

9. Ans (2)



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B)

$$f\left(\frac{\pi^-}{2}\right) = f\left(\frac{\pi^+}{2}\right) = f\left(\frac{\pi}{2}\right)$$

$$\Rightarrow 1 = 1 + b \Rightarrow b = 0$$

$$\text{Now, } f'\left(\frac{\pi^-}{2}\right) = f'\left(\frac{\pi^+}{2}\right)$$

$$\Rightarrow 1 = 0 + 0 + c(-1)(2)$$

$$\Rightarrow c = -\frac{1}{2}$$

2. Ans (C)

Homogeneous equation

$$\Rightarrow \frac{d^2y}{dx^2} = 0$$

3. Ans (D)

$$f(0) = f(1)$$

$$0 = 1 + 1 + a + b$$

$$a + b = -2 \quad \dots(1)$$

$$\& f'\left(\frac{1}{4}\right) = 0 \Rightarrow \frac{4}{64} + \frac{3}{16} + \frac{2a}{4} + b = 0$$

$$\Rightarrow 1 + 2a + 4b = 0 \quad \dots(2)$$

$$\text{from (1) \& (2) } b = \frac{3}{2}$$

$$a = -\frac{7}{2}$$

4. Ans (D)

(1, 2) lies on y_1

$$1 = a + b + \frac{7}{2} \Rightarrow a + b = -\frac{3}{2} \quad \dots(1)$$

$$\left(\frac{dy_1}{dx}\right)_{(1,2)} = -\frac{1}{\left(\frac{dy}{dx}\right)_{(-2,2)}}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,B,D)

$$(A) e^x \geq 1 \Rightarrow x \geq 0$$

$$\text{and } e^x - 1 > -1$$

$$(B) \sqrt{2} < \cot^{-1}x < 4$$

$$\Rightarrow -\infty < x < \cot \sqrt{2}$$

(D) domain different $\therefore$ not identical range

6. Ans (C,D)

$$\text{Let } \frac{\pi}{5^x} = t$$

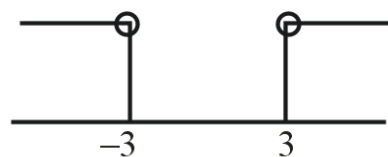
$$\Rightarrow \lim_{t \rightarrow 0} \frac{\pi}{5^t} (\sin t + \tan t)$$

$$= \frac{2\pi}{5}$$

7. Ans (A,D)

$$9 - x^2 < 0 \quad \text{and} \quad -8x < 0$$

$$x^2 - 9 > 0 \quad x > 0$$



$$9 - x^2 > -8x$$

$$x^2 - 8x - 9 < 0$$

$$(x - 9)(x + 1) < 0$$

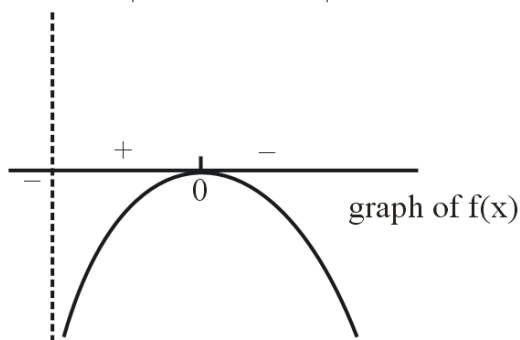
$$\Rightarrow x \in (-1, 9)$$

$$\Rightarrow x \in (3, 9)$$

8. Ans (A,C,D)

Consider $f(x) = \ln(1+x) - x$

$$f'(x) = \frac{1}{1+x} - 1 = -\frac{x}{1+x}$$



$$f(x) < 0 \quad \forall x \in (-1, \infty) - \{0\}$$

9. Ans (A,B,C,D)

$$f'(x) = \begin{vmatrix} 6x+2 & 10x+4 & 14x+6 \\ 3x+1 & 5x+2 & 7x+3 \\ 3 & 5 & 7 \end{vmatrix} + \begin{vmatrix} 3x^2+2x & 5x^2+4x+3 & 7x^2+6x+5 \\ 3 & 5 & 7 \\ 3x^2+2x & 5x^2+4x+3 & 7x^2+6x+5 \end{vmatrix} + \begin{vmatrix} 3x+1 & 5x+2 & 7x+3 \\ 0 & 0 & 0 \end{vmatrix} = 0$$

$$f'(x) = 0 \Rightarrow f(x) = k \quad f(0) = 1$$

$$\Rightarrow f(x) = 1 \quad \text{check options.}$$

10. Ans (B,C,D)

$$y = \frac{x+1}{x^2+1}$$

$$\frac{d^2y}{dx^2} = \frac{2x^3 + 6x^2 - 6x - 2}{(x^2+1)^3}$$

$$\Rightarrow 2\alpha^3 + 6\alpha^2 - 6\alpha - 2 = 0$$

$$\alpha^3 + 3\alpha^2 - 3\alpha - 1 = 0$$

$$\alpha = 1, -2 - \sqrt{3}, -2 + \sqrt{3}$$

$$\Pi\alpha = 1, \Sigma\alpha = -3$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (0.00)

$$\lim_{\theta \rightarrow \frac{\pi}{2}} \sqrt{\sec \theta} - \sqrt{\tan \theta} = \lim_{\theta \rightarrow \frac{\pi}{2}} \frac{1 - \sqrt{\sin \theta}}{\sqrt{\cos \theta}} = \lim_{\theta \rightarrow \frac{\pi}{2}} \left(\frac{\cos \theta}{\sin \theta} \right)^{3/2} = 0$$

(L Hopital Rule)

2. Ans (0.00)

$$\lim_{\theta \rightarrow \frac{\pi}{2}} \frac{1 - \sin \theta}{\cos \theta} = \lim_{\theta \rightarrow \frac{\pi}{2}} \frac{\cos \theta}{\sin \theta} = 0$$

(L Hopital Rule)

3. Ans (1.00)

$$g(x) = x + \ln x + c$$

$$g(1) = 1 + c = 1 \Rightarrow c = 0$$

$$\Rightarrow g(x) = x + \ln x$$

$$\Rightarrow f(x) = (x + \ln x) e^x$$

$$\lim_{x \rightarrow \infty} \frac{(x + \ln x) e^{x^2}}{x e^{x^2}} = \lim_{x \rightarrow \infty} \frac{1 + \ln x}{x} = 1$$

4. Ans (0.25)

$$\phi'(f(x)) = \frac{1}{f'(x)}$$

$$\text{put } f(x) = e \Rightarrow x = 1$$

$$\Rightarrow \phi'(e) = \frac{1}{f'(1)} = \frac{1}{4e}$$

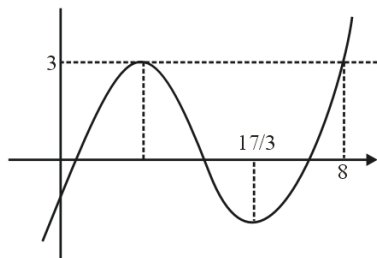
5. **Ans (3.00)**

$$f(x) = 3x^2 - 20x + 17$$

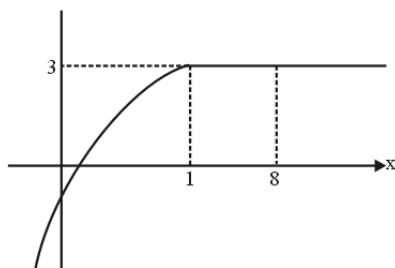
$$= 3x^2 - 3x - 17x + 17$$

$$= (3x - 17)(x - 1)$$

graph of $f(x)$



graph of $g(x)$



clearly $g(x) = 3$ has infinite solutions

Range of $g(x)$ is $(-\infty, 3]$

$g(x)$ is non differentiable at only $x = 1$

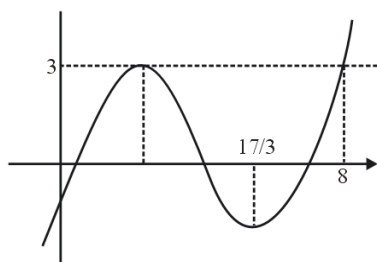
6. **Ans (0.33 or 0.34)**

$$f(x) = 3x^2 - 20x + 17$$

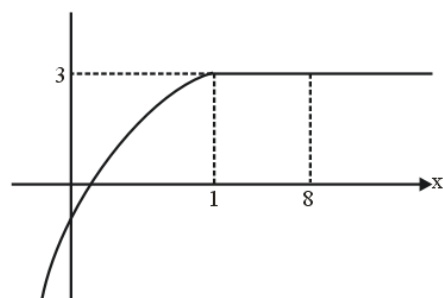
$$= 3x^2 - 3x - 17x + 17$$

$$= (3x - 17)(x - 1)$$

graph of $f(x)$



graph of $g(x)$



clearly $g(x) = 3$ has infinite solutions

Range of $g(x)$ is $(-\infty, 3]$

$g(x)$ is non differentiable at only $x = 1$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. **Ans (1)**

We have

$$f(x) = \left(4\cos^4 x - 2(2\cos^2 x - 1) - \frac{1}{2}(2\cos^2 2x - 1) - x^7 \right)^{\frac{1}{7}}$$

$$\Rightarrow f(x) = \left(\frac{3}{2} - x^7 \right)^{\frac{1}{7}}$$

$$f(f(x)) = \left[\frac{3}{2} - (f(x))^7 \right]^{\frac{1}{7}} = x$$

$$\Rightarrow f'(f(x)) \times f'(x) = 1$$

8. **Ans (14)**

$$\int (\cos^2 x - \sin^2 x) (\cos^4 x + \sin^4 x) dx$$

$$= \int (\cos 2x) \left(1 - \frac{1}{2} \sin^2 2x \right) dx$$

$$t = \sin 2x$$

$$= \frac{1}{4} \int (2 - t^2) dt = \frac{1}{4} \left(2t - \frac{t^3}{3} \right) + c$$

$$\Rightarrow a = \frac{1}{2}, b = \frac{1}{12}$$

$$\Rightarrow \frac{1}{a} = 2, \frac{1}{b} = 12$$

9. **Ans (2)**

$$f'(x) = 4x^3 + 18x^2 + 70x + 6$$

$$\text{and } f''(x) + 12x^2 + 36x + 70$$

$$= 12 \left(x^2 + 3x + \frac{9}{4} \right) + 43$$

$$\therefore f''(x) = 12 \left(x + \frac{3}{2} \right)^2 + 43 > 0 \forall x \in \mathbb{R}$$

so $f(x)$ has no inflection point and

$f'(x) = 4x^3 + 18x^2 + 70x + 6$ is strictly increasing

function, so equation

$f'(x) = 0$ has exactly one real root

$\Rightarrow f(x)$ has 1 stationary point.

Since $f(x)$ is concave up $\forall x \in \mathbb{R}$

$\Rightarrow$ stationary point is the point of local minima

$$\therefore m = 0, n = 1, p = 1$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

REVIEW TEST

01-05-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & LIVE-I(A)]

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B	A,B	A,B,D	B,C	A,C	A
SECTION-I (ii)	Q.	7	8	9	10		
	A.	B	B	A	D		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	2.00	4.00	-46.00	19.67	2.00	4.00
SECTION-II (ii)	Q.	7	8	9			
	A.	6	4	3			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B,C,D	A,B,C,D	D	A,B	B,C,D	B,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	C	D	A	D		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	3.00	2.00	0.56	1.25	0.33	2.00
SECTION-II (ii)	Q.	7	8	9			
	A.	3	3	0			

PART-3 : MATHEMATICS

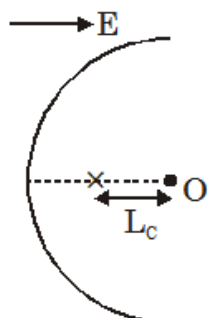
SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	C,D	B,C,D	A,C	B,C	A,C,D	A,B,C,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	C	B	B	B		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	3.50	0.00	3.00	2.00	8.00	2.00
SECTION-II (ii)	Q.	7	8	9			
	A.	3	6	0			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,B)



$$T = 2\pi \sqrt{\frac{I}{mg \cdot L_c}}$$

$$\text{Put } I = mR^2$$

$$g = \frac{\lambda (\pi R) E}{m}$$

2. Ans (A,B)

A = deformation in equilibrium state.

$$2kA = \frac{\sigma}{\epsilon_0} \cdot q$$

$$\therefore A = \frac{\sigma q}{2k\epsilon_0}$$

Springs are connected in parallel $K_{eq} = 2k$

$$\text{Angular frequency} = \sqrt{\frac{2k}{m}}$$

4. Ans (B,C)

By energy conservation $\frac{1}{2}kx^2 = mg(2x)$

$$\Rightarrow x = \frac{4mg}{k} \text{ (maximum elongation)}$$

$$\text{at equilibrium } kx = 2mg \Rightarrow x = \frac{2mg}{k}$$

$$\text{So, (K.E.)}_{\max} = mg(2x) - \frac{1}{2}kx^2$$

$$= 2mg \left(\frac{2mg}{k} \right) - \frac{1}{2}k \left(\frac{2mg}{k} \right)^2$$

$$\frac{1}{2}mv_{\max}^2 = \frac{2m^2g^2}{k}$$

$$\Rightarrow v_{\max} = 2g\sqrt{\frac{m}{k}}$$

5. Ans (A,C)

$$\frac{\mu_1}{v} - \frac{\mu_2}{u} = \frac{\mu_1 - \mu_2}{R}$$

$$\Rightarrow \frac{\mu_1}{v} - \frac{\mu_2}{u} < 0$$

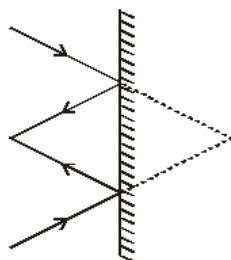
$$\Rightarrow \frac{\mu_1}{v} < \frac{\mu_2}{u}$$

$$\Rightarrow \frac{v}{u} > \frac{\mu_1}{\mu_2}$$

$$\Rightarrow \frac{v}{u} > 0$$

$\Rightarrow v$ and u must have same sign.

6. Ans (A)

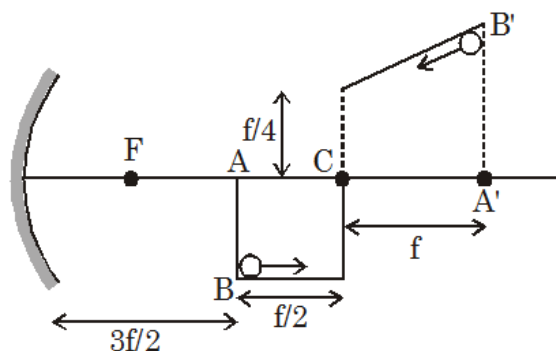


PART-1 : PHYSICS

SECTION-I (ii)

9. Ans (A)

The image of the object can be easily drawn by using the mirror formula. Right upright side of wire frame is on centre of curvature so its image would be real, inverted and of same size forming at the centre of curvature. For left upright side i.e. for point A. Use mirror formula.



$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{v} - \frac{1}{\frac{3f}{2}} = -\frac{1}{f} \Rightarrow v = -3f$$

i.e. image of A forms at $-3f$ distance from pole.

Height of image of AB can be computed by

using magnification formula $m = \frac{h_1}{h_0} = -\frac{v}{u}$

$$\Rightarrow h_i = - \left[\frac{-3f}{-\frac{3f}{2}} \right] \times \left[-\frac{f}{4} \right] = \frac{f}{2}$$

The top points of upright section of image of wire frame would be joined by a straight line, as the image of linear object would be always straight.

The particle always remain in contact with the lower edge of wire frame so the image of it moves along the slant part of the image.

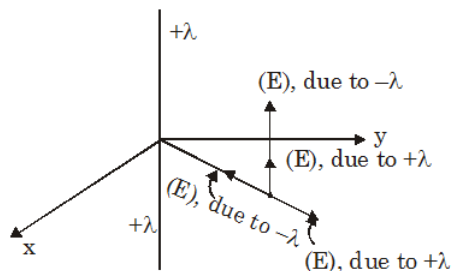
10. Ans (D)

It is clear from ray diagram that the velocity of image of particle is varying.

PART-1 : PHYSICS

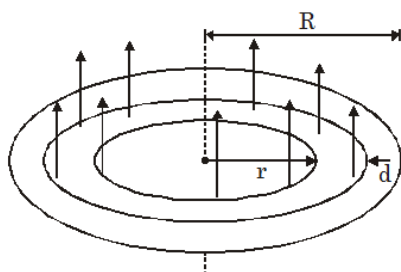
SECTION-II (i)

1. Ans (2.00)



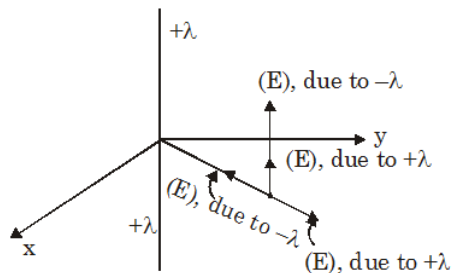
$$E_{\text{net}} = \frac{2k\lambda}{r} = \frac{\lambda}{2\pi\epsilon_0 r}$$

$$\phi_{\text{Net}} = \int_0^R \frac{\lambda}{2\pi\epsilon_0 r} \cdot 2\pi r \cdot dr$$



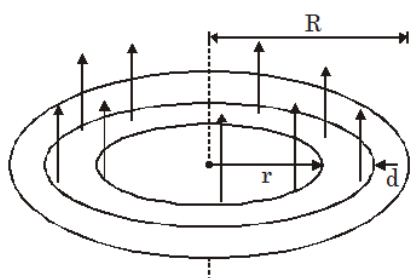
$$\phi_{\text{Net}} = \frac{\lambda}{\epsilon_0} R$$

2. Ans (4.00)



$$E_{\text{net}} = \frac{2k\lambda}{r} = \frac{\lambda}{2\pi\epsilon_0 r}$$

$$\phi_{\text{Net}} = \int_0^R \frac{\lambda}{2\pi\epsilon_0 r} \cdot 2\pi r \cdot dr$$



$$\phi_{\text{Net}} = \frac{\lambda}{\epsilon_0} R$$

3. Ans (-46)

$$\vec{p}_i = 1(8\hat{i} + 6\hat{j}) + 2(5\hat{i} - 5\hat{j}) = 18\hat{i} - 4\hat{j}$$

$$\vec{p}_i = \vec{f}_{\text{ext}} \Delta t + \vec{p}_{i1} = -60\hat{j} + 18\hat{i} - 4\hat{j} = 18\hat{i} - 64\hat{j}$$

4. Ans (19.67)

Initial velocity of centre of mass

$$u_{\text{cm}} = \frac{\vec{p}_i}{m_1 + m_2} = \frac{18\hat{i} - 4\hat{j}}{3} = 6\hat{i} - \frac{4}{3}\hat{j}$$

$$\therefore \vec{r}_{\text{cm}(t=1)} = \vec{r}_{\text{cm}(t=0)} + \vec{u}_{\text{cm}}(1) + \frac{1}{2}\vec{a}_{\text{cm}}(1)^2$$

$$= 20\hat{j} + \left(6\hat{i} - \frac{4}{3}\hat{j}\right) + \frac{1}{2}(-10\hat{j}) = 6\hat{i} + \frac{41}{3}\hat{j}$$

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (6)

$$U_i + k_i = U_f + k_f$$

$$-\frac{kq^2}{5^2} + 1.5 = -\frac{kq^2}{r^2} + 0$$

$$r = 6\text{m}$$

8. Ans (4)

$$U = 2\left(-\frac{kQ^2}{r}\right) + \frac{kQ^2}{2r}$$

$$= \frac{kQ^2}{r}\left(-2 + \frac{1}{2}\right) = -\frac{3}{2}\left(\frac{kQ^2}{r}\right)$$

$$\frac{kQ^2}{r^2} - \frac{kQ^2}{4r^2} = \frac{mv^2}{r}$$

$$\frac{3kQ^2}{4r^2} = \frac{mv^2}{r} \Rightarrow mv^2 = \frac{3kQ^2}{4r}$$

$$\text{T.E.} = -\frac{3}{2}\frac{kQ^2}{r} + \frac{3kQ^2}{4r}$$

$$= \frac{kQ^2}{r}\left(-\frac{3}{2} + \frac{3}{4}\right) = \frac{kQ^2}{r}\left(-\frac{3}{4}\right)$$

9. Ans (3)

$$F = 10 \times 12 = 120 \text{ N}$$

$$F = kx = 2400 x$$

$$\therefore x = \frac{1}{20}$$

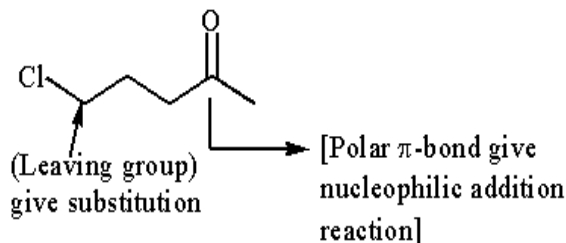
$$\text{Energy stored in the spring } E = \frac{1}{2}kx^2$$

$$= \frac{1}{2} \times 2400 \times \frac{1}{400} = 3\text{J}$$

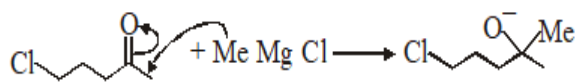
PART-2 : CHEMISTRY

SECTION-I (i)

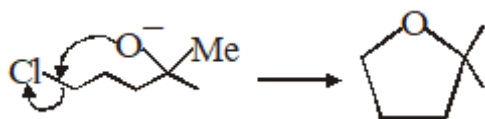
2. Ans (A,B,C,D)



(i) Grignard prefer to give nucleophilic addition on polar π -bond and from anion intermediate.



(ii) In next step anion give intramolecular nucleophilic substitution reaction & form 5 membered ring.



4. Ans (A,B)

$\Rightarrow$ extent of synergic bonding increases as the $-ve$ charge increases on metal

$\Rightarrow$ when synergic bonding is high, M-C bond length decreases, C-O bond length increases.

5. Ans (B,C,D)

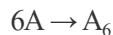
B, C & D options are correct

PART-2 : CHEMISTRY

SECTION-I (ii)

10. Ans (D)

$$n_A = 10; n_B = 20; n_C = 5$$



$$-\frac{1}{6} \frac{d[A]}{dt} = k = 0.6; -\frac{dA}{dt} = 0.6 \times \frac{1}{6} = 0.1$$

$$At = A_0 - k \times t$$

$$= 10 - 0.1 \times 60 = 4 \text{ moles of } A_2$$

$$\text{now 4 moles of Aliq.} = 20; n_C = 5$$

$$P(s) = \frac{4}{30} \times 100 + \frac{20}{30} \times 90 = \frac{40 + 180}{3} = \frac{220}{3}$$

After 100 min 'A' will finish and convert to A_6

$$\eta_A = \frac{10}{6} = \frac{5}{3}; \eta_A = 20; \eta_A = 5$$

$$\eta_T = \frac{5}{3} = 20 + 5 \Rightarrow \frac{5 + 60 + 15}{3} = \frac{80}{3}$$

$$P(s) = \frac{5/3 \times 100 + 20 \times 90}{80/30}$$

$$= \frac{5800}{80} = 73.75$$

PART-2 : CHEMISTRY

SECTION-II (i)

1. Ans (3.00)

(ii), (iii) (iv)

2. Ans (2.00)

(i), (v)

4. Ans (1.25)

$$-\frac{d[A]}{dt} = \frac{1}{2} \frac{d[B]}{dt} = k[A]$$

$$\text{at } t = \infty; [B] = 2M$$

$$\text{so } A_0 = 1M$$

$$\text{when } [B] = 1.2$$

$$[A] = 0.4 \text{ \& } \frac{d[B]}{dt} = 1$$

$$\text{so } \frac{d[A]}{dt} = \frac{1}{2} = K \times (0.4)$$

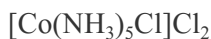
$$\frac{1}{2} = K \times \frac{2}{5}$$

$$K = \frac{5}{4}$$

$$t_{1/2} = \frac{0.7 \times 5}{5} = 0.56$$

$$r = K \times A = \frac{5}{4} = 1.25$$

5. **Ans (0.33)**



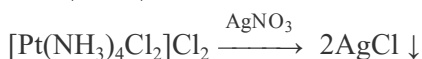
$$x = 3$$

$$y = 6$$

$$z = 3$$

$$\Rightarrow \frac{3}{3+6} = 0.33$$

6. **Ans (2.00)**



PART-2 : CHEMISTRY

SECTION-II (ii)

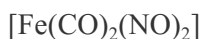
8. **Ans (3)**



9. **Ans (0)**



$$\text{EAN} = 26 - 0 + 10 = 36$$



$$\text{EAN} \Rightarrow 26 - 0 + 2 \times 2 + 2 \times 3 = 36$$

$$\text{Difference} \Rightarrow 0$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (C,D)**

Let tangent at (t^2, t^3) passes through $\left(0, -\frac{1}{2}\right)$

$$2y \frac{dy}{dx} = 3x^2 \Rightarrow \frac{dy}{dx} = \frac{3x^2}{2y} = \frac{3t}{2}$$

$$\frac{dy}{dx} = \frac{y_2 - y_1}{x_2 - x_1} \Rightarrow \frac{t^3 + \frac{1}{2}}{t^2} = \frac{3t}{2}$$

$$\Rightarrow 2t^3 + 1 = 3t^3 \Rightarrow t = 1$$

$$\Rightarrow \text{Equation of tangent is } y - 1 = \frac{3}{2}(x - 1)$$

$$\Rightarrow 2y - 2 = 3x - 3 \Rightarrow 3x = 2y + 1$$

2. **Ans (B,C,D)**

$$f(x) = \cos^{-1}([e^x] - 1) + \sin^{-1}([e^x])$$

$$D_f : [e^x] \in [0, 1] \Rightarrow [e^x] = 0 \text{ or } [e^x] = 1$$

$$\text{hence, } x \in (-\infty, \ln 2) \text{ and } f(x) = \pi$$

3. **Ans (A,C)**

$$L = \lim_{x \rightarrow 0} \frac{(x - \sin x)}{x^3} \cdot \frac{(x^{n-1} + x^{n-2} \sin x + \dots + (\sin x)^{n-1})}{x^{n-1}} \cdot \frac{1}{x^{m-n-2}}$$

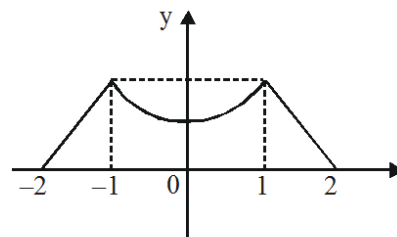
$$\therefore L = \frac{1}{6} \cdot n \cdot \lim_{x \rightarrow 0} \frac{1}{x^{m-n-2}}$$

$$\text{at } m - n = 2 : L = \frac{n}{6}$$

if $m - n > 2$: L does not exist

if $m - n < 2$: L = 0

4. **Ans (B,C)**



Non derivable at
-1, 0, 1

Graph of $f(x)$ is shown in one of its periodic interval. Range of $f(x)$: $[0, e]$

$f(x)$ is continuous at all points and non derivable at 19 points.

5. **Ans (A,C,D)**

Function is continuous in $[0, 2]$

$$\Rightarrow 3 = a \quad \dots\dots(i)$$

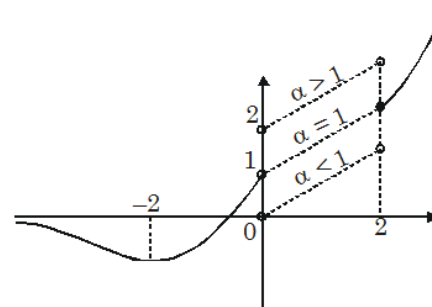
$$-1 + a = m + b \quad \dots\dots(ii)$$

function is differentiable in $(0, 2)$

$$\Rightarrow -2 = m \quad \dots\dots(iii)$$

6. **Ans (A,B,C,D)**

$$f(x) = \begin{cases} (x+1)e^x & x \leq 0 \\ \frac{x}{2} + a & 0 < x < 2 \\ x^2 - 2 & x \geq 2 \end{cases}$$



PART-3 : MATHEMATICS

SECTION-I (ii)

7. **Ans (C)**

$$f(xy) = f(x) + f(y) \quad \forall x, y > 0.$$

$$\text{then } f(x) = k \ln(x)$$

$$f(x) = k \cdot \frac{1}{x}$$

$$f'(1) = k = 3$$

$$\therefore f(x) = 3 \ln x$$

$f(x)$ is differentiable for all $x \in \mathbb{R}^+$

8. **Ans (B)**

$$f(xy) = f(x) + f(y) \quad \forall x, y > 0.$$

$$\text{then } f(x) = k \ln(x)$$

$$f(x) = k \cdot \frac{1}{x}$$

$$f'(1) = k = 3$$

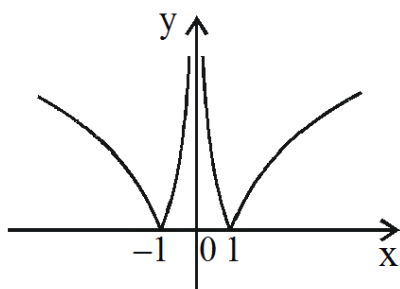
$$\therefore f(x) = 3 \ln x$$

$$y = |f(|x|)| = |3 \ln(|x|)|$$

Domain : $\mathbb{R}_0$

Continuous at all points in domain.

Not differentiable at $x = -1$ and 1



9. **Ans (B)**

$$f^{-1}(x) = xe^x = g(x) \text{ (say)} \Rightarrow f(x) = g^{-1}(x)$$

$$\text{clearly, } g(e) = e^{(e+1)} \Rightarrow g^{-1}(e^{(e+1)}) = e$$

$$\Rightarrow f(e^{(e+1)}) = e \Rightarrow f(f(e^{(e+1)})) = f(e)$$

$$= g^{-1}(e) = 1$$

$$\text{Now, } ye^y = x; y = f(x)$$

$$\text{So, } \ln y + y = \ln x \Rightarrow \frac{y'}{y} + y' = \frac{1}{x}$$

$$\Rightarrow f'(x) = \frac{y}{x(y+1)}$$

$$\therefore \lim_{x \rightarrow \infty} \frac{f(x)}{\ln x} = \lim_{x \rightarrow \infty} x f'(x) = \lim_{x \rightarrow \infty} \frac{f(x)}{f(x)+1} = 1$$

(as $f(x) \rightarrow \infty$ when $x \rightarrow \infty$)

10. **Ans (B)**

$$f^{-1}(x) = xe^x = g(x) \text{ (say)} \Rightarrow f(x) = g^{-1}(x)$$

$$\text{clearly, } g(e) = e^{(e+1)} \Rightarrow g^{-1}(e^{(e+1)}) = e$$

$$\Rightarrow f(e^{(e+1)}) = e \Rightarrow f(f(e^{(e+1)})) = f(e)$$

$$= g^{-1}(e) = 1$$

$$\text{Now, } ye^y = x; y = f(x)$$

$$\text{So, } \ln y + y = \ln x \Rightarrow \frac{y'}{y} + y' = \frac{1}{x}$$

$$\Rightarrow f'(x) = \frac{y}{x(y+1)}$$

$$\therefore \lim_{x \rightarrow \infty} \frac{f(x)}{\ln x} = \lim_{x \rightarrow \infty} x f'(x) = \lim_{x \rightarrow \infty} \frac{f(x)}{f(x)+1} = 1$$

(as $f(x) \rightarrow \infty$ when $x \rightarrow \infty$)

PART-3 : MATHEMATICS

SECTION-II (i)

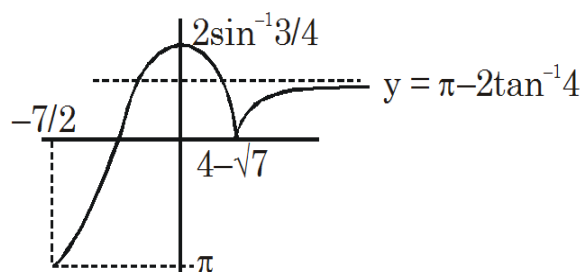
1. **Ans (3.50)**

$$\text{at } x = 0 \Rightarrow (2a-1) \sin^{-1} \frac{3}{4} = b \cos^{-1} \frac{\sqrt{7}}{4} \Rightarrow 2a-1 = b$$

$$\text{at } x = 4 - \sqrt{7} \Rightarrow b \tan^{-1} A - 2 \tan^{-1} 4 = 0 \Rightarrow b = 2$$

$$a = \frac{3}{2}, b = 2$$

range of $f(x)$ is $[-\pi, \pi - 2 \tan^{-1} 4]$



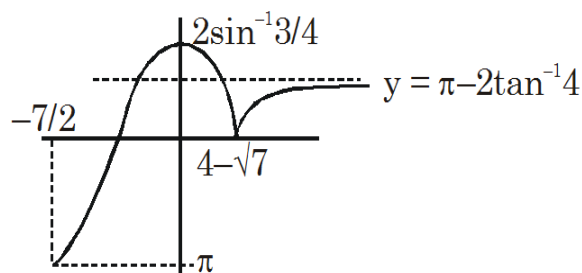
2. **Ans (0.00)**

$$\text{at } x = 0 \Rightarrow (2a-1) \sin^{-1} \frac{3}{4} = b \cos^{-1} \frac{\sqrt{7}}{4} \Rightarrow 2a-1 = b$$

$$\text{at } x = 4 - \sqrt{7} \Rightarrow b \tan^{-1} A - 2 \tan^{-1} 4 = 0 \Rightarrow b = 2$$

$$a = \frac{3}{2}, b = 2$$

range of $f(x)$ is $[-\pi, \pi - 2 \tan^{-1} 4]$



3. **Ans (3.00)**

$$f(x) = x^3 - 6x^2 + 9x + 1$$

∴ Rolle's theorem holds in $[0, a]$

$$\therefore f(0) = f(a)$$

$$\Rightarrow 1 = a^3 - 6a^2 + 9a + 1$$

$$\Rightarrow a(a^2 - 6a + 9) = 0$$

$$\Rightarrow a(a - 3)^2 = 0$$

$$\Rightarrow a = 0, 3$$

4. **Ans (2.00)**

$$f(x) = 3x^2 - 12x + 9$$

$$f'(x) = 6x - 12 = 0$$

$$x = 2$$

5. **Ans (8.00)**

$$I = \int \frac{\cos^4 x \cdot \cos x}{\sin x} \cdot dx$$

put $\sin x = t$

$$= \int \frac{(1 - t^2)^2 dt}{t} = \int \frac{(1 + t^4 - 2t^2)}{t} dt$$

$$= \log|t| + \frac{t^4}{4} - t^2 + C$$

$$= \log|\sin x| + \frac{1}{4} \cdot \sin^4 x - \sin^2 x + C$$

$$\frac{\ell - n}{m} = \frac{2}{\frac{1}{4}} = 8$$

6. **Ans (2.00)**

$$I = \int \frac{\cos^4 x \cdot \cos x}{\sin x} \cdot dx$$

put $\sin x = t$

$$= \int \frac{(1 - t^2)^2 dt}{t} = \int \frac{(1 + t^4 - 2t^2)}{t} dt$$

$$= \log|t| + \frac{t^4}{4} - t^2 + C$$

$$= \log|\sin x| + \frac{1}{4} \cdot \sin^4 x - \sin^2 x + C$$

$$\frac{\ell - n}{m} = \frac{2}{\frac{1}{4}} = 8$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. **Ans (3)**

$$x - y + z = 1$$

$$\lambda x + 3y + 2z = 3$$

$$3x + \lambda y + z = 2$$

$$\Delta = \begin{vmatrix} 1 & -1 & 1 \\ \lambda & 3 & 2 \\ 3 & \lambda & 1 \end{vmatrix} \neq 0$$

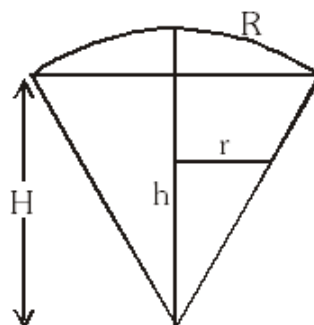
$$\Rightarrow (3 - 2\lambda) + 1(\lambda - 6) + 1(\lambda^2 - 9) \neq 0$$

$$\Rightarrow \lambda^2 - 9 + \lambda - 6 + 3 - 2\lambda \neq 0$$

$$\Rightarrow \lambda^2 - \lambda - 12 \neq 0 \Rightarrow \lambda \neq 4, -3$$

$$4 \left| \frac{k_2}{k_1} \right| = 4 \left| \frac{3}{4} \right| = 3$$

8. **Ans (6)**



$$\frac{r}{h} = \frac{R}{H} \Rightarrow r = \frac{h}{2}$$

$$v = \frac{1}{3} \pi r^2 h = \frac{1}{3} \pi \frac{h^3}{4} \Rightarrow \frac{dv}{dt} = \frac{\pi h^2}{4} \frac{dh}{dt}$$

$$\Rightarrow 23100 = \frac{22}{7} \times \frac{4900 \times dh/dt}{4}$$

$$\Rightarrow \frac{dh}{dt} = 6 \text{ cm/min.}$$

9. **Ans (0)**

$$f'(x) = \frac{(1+x) \cdot \frac{1}{x} - (\ln x) \cdot 1}{(1+x)^2} - 1$$

$$= \underbrace{\frac{1-x(1+x)}{x(1+x)}}_{\text{negative}} - \underbrace{\frac{\ln x}{(1+x)^2}}_{\text{positive}} < 0 \text{ for } x \geq 1$$

Therefore $f(x)$ has no critical point.